# Week 9 Notes & Quiz

Notes for all three Week 9 lectures, made only from the 93 slides in *WEEK 9.pdf*. Each yellow tag such as S49 is the slide (PDF page) the point comes from. The 41-question practice questions and answer key are at the end.

| STOP | LECTURE | SLIDES |
|---|---|---|
| L42 | **Importance of Distribution in Supply Chain** | S1–S25 |
| L43 | **Designing Distribution Networks in Supply Chain** | S26–S57 |
| L44 | **Facility Location Planning** | S58–S93 |
| + | **Cheat sheet · Slide quirks · Quiz (41 Qs)** | — |

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LECTURE 42 S1–S25

## Importance of Distribution in Supply Chain

> DISTRIBUTION S3
>
> The steps taken to **move and store a product** from the supplier stage to a customer stage. It is a key driver of profitability because it affects both **supply chain cost and customer value**. Walmart uses distribution to offer high availability of common products at very low cost.

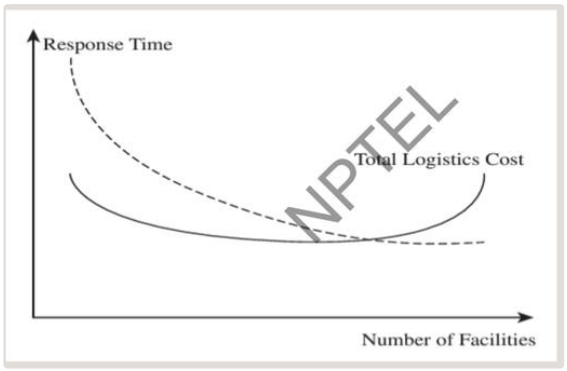
- Network design has two phases: (1) the broad structure, e.g. sell direct or through an intermediary; (2) specific locations, capabilities, capacities and demand allocation. S4

### Service factors 7 items

**Response time** (short response → facilities close to customers) · **product variety** · **product availability** (probability of being in stock) · **customer experience** · **time to market** · **order visibility** · **returnability**. S5–S6

### Cost factors

- Changing the network affects **inventories, transportation, facilities and handling, information**. Sourcing and pricing also matter. S7
- Shorter desired response time → more facilities needed. S8
- More facilities → **inventory costs rise** S9; **facility costs rise** S12; **transport costs fall** until inbound economies of scale are lost, then rise (U-shape) S10.
- **Inbound** transport = bringing material into a facility; **outbound** = sending it out. Outbound cost per unit is usually higher because inbound lots are larger (Amazon receives full truckloads of books, ships small parcels). S11




Original slide S13. As facilities increase, **response time falls** and **total logistics cost** first falls then rises.

### Design options

Direct shipping · centralized distribution · decentralized distribution · cross-docking · **drop-shipping** (suppliers ship directly to customers, no inventory held) · hybrid · 3PL providers. S14–S15

### Online sales and distribution

- Considerations: fulfilment centres, inventory management, last-mile delivery, order processing and packaging, returns, technology, customer experience, scalability, integration with stores (ship-from-store, in-store pickup), data analytics. S16–S18
- Effect on service: physical products take longer (digital ones are instant); more **variety** (Netflix vs rental stores); better availability via aggregation; 24/7 access; faster time to market (Walmart.com); order visibility; returns are **harder and costlier**; direct sales; flexible pricing; efficient funds transfer. S19–S20

| COST | EFFECT OF ONLINE SALES S21–S23 |
|---|---|
| **Inventory** | **Lower** via aggregation (Amazon vs Barnes & Noble) and **postponement** (Dell assembles after order; Amazon prints rare books on demand). |
| **Facilities** | Fewer facilities: **Netflix ~50 warehouses** vs Blockbuster's thousands of stores. But groceries cost more to handle online. |
| **Transportation** | Digital goods: far cheaper. Physical goods: **outbound cost rises** with centralized inventory. |

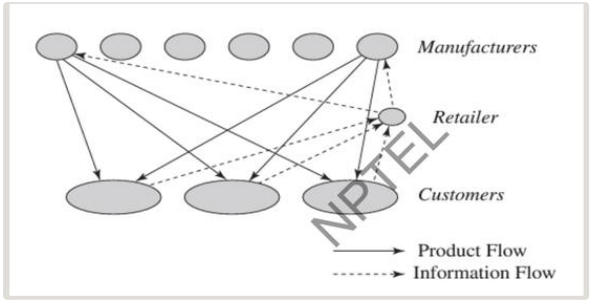
| COST | EFFECT OF ONLINE SALES S21–S23 |
|---|---|
| **Information** | Better visibility, but infrastructure costs (Zappos: about 120,000 descriptions, 2 million+ photos). |

LECTURE 43 S26–S57

# Designing Distribution Networks

## Six distribution network designs S28

| DESIGN | HOW IT WORKS · HIGHLIGHTS |
|---|---|
| **1. Manufacturer storage with direct shipping (drop-shipping)** | Manufacturer holds stock and ships to the customer; retailer passes orders on. Lowest inventory (aggregation), high transport cost, response time **one to two weeks**, easy variety, returns expensive. S29–S32 |
| **2. … with in-transit merge** | Parts from several sources are combined by the carrier so the customer gets **one delivery** (e.g. PC + monitor). Somewhat lower transport than drop-shipping; better customer experience. S33–S35 |
| **3. Distributor storage with carrier delivery** | Distributor/retailer warehouses hold stock; package carriers deliver. Faster response, higher inventory than manufacturer storage. S36–S38 |
| **4. Distributor storage with last-mile delivery** | Distributor delivers to the customer's home; very quick (same/next day), **highest transport cost**; grocery and e-commerce. S39–S42 |
| **5. Manufacturer/distributor storage with customer pickup** | Customer collects from a pickup site: lower transport, limited convenience. S43–S45 |
| **6. Retail storage with customer pickup** | Stock in local stores; same-day pickup, inspect before buying; **highest inventory and facility costs**, lowest transport. S46–S47 |




Original slide S30, drop-shipping: product flows from manufacturers straight to customers (solid lines); the retailer handles only the information flow (dashed).

Comparative Performance of Delivery Network Designs

| | Retail Storage with Customer Pickup | Manufacturer Storage with Direct Shipping | Manufacturer Storage with In-Transit Merge | Distributor Storage with Package Carrier Delivery | Distributor Storage with Last-Mile Delivery | Manufacturer Storage with Pickup |
|---|---|---|---|---|---|---|
| Response time | 1 | 4 | 4 | 3 | 2 | 4 |
| Product variety | 4 | 1 | 1 | 2 | 3 | 1 |
| Product availability | 4 | 1 | 1 | 2 | 3 | 1 |
| Customer experience | Varies from 1 to 5 | 4 | 3 | 2 | 1 | 5 |
| Time to market | 4 | 1 | 1 | 2 | 3 | 1 |
| Order visibility | 1 | 5 | 4 | 3 | 2 | 6 |
| Returnability | 1 | 5 | 5 | 4 | 3 | 2 |
| Inventory | 4 | 1 | 1 | 2 | 3 | 1 |
| Transportation | 1 | 4 | 3 | 2 | 5 | 1 |
| Facility and handling | 6 | 1 | 2 | 3 | 4 | 5 |
| Information | 1 | 4 | 4 | 3 | 2 | 5 |

Original slide S49 (Chopra & Meindl). **1 = strongest, 6 = weakest.** Retail storage with pickup is best on response time, order visibility, returnability, transportation and information, but worst (6) on facility and handling. Manufacturer storage designs are best (1) on variety, availability, time to market and inventory.

- Selecting a design: customer needs, product characteristics (value, demand variability, perishability), costs, industry dynamics, scalability, ownership, overall strategy. S48
- Suitability (S50): drop-shipping suits **low and very-low demand**, high-value, high-variety products (+2); retail pickup suits **high-demand** products and quick response (+2). Key: +2 very suitable … −2 very unsuitable. S50

## Case: Dell's online sales

- Dell pioneered online sales of computer hardware (about half the category by 2009) and began selling via Wal-Mart in **2007**. S51
- Benefits: customization, fast time to market, 24/7 low-cost service, **negative working capital** (paid before it pays suppliers), inventory reduction via aggregation and **postponement**. Drawback: higher outbound transport (single PCs shipped). S52–S53
- Online works well for **high-value customized** PCs; less so for **low-cost standardized** PCs (response time −2, transport −2). S53–S54
- Conclusion: hybrid online + offline models are common; Amazon and Blue Nile profited, **Webvan** struggled. S55

LECTURE 44 S58–S93

# Facility Location Planning

- The facility location problem (FLP, location analysis) is a branch of operations research and computational geometry: place facilities to minimize transport cost, considering e.g. keeping hazardous materials away from housing. S60
- Location decisions are **strategic**: they affect the whole organization, last long, are **difficult to reverse** and **capital intensive**; new plants, expansion, decentralization and shutdowns are constantly reviewed. S60
- If a location was poorly planned: sell it (disinvestment), close and liquidate, continue operating, or relocate. S61–S62
- **Three-tier model** of location competitiveness: **country** (government rules, openness to trade, finance), **sector** (efficiency in management, competitive design and marketing), **company** (skilled labour, infrastructure, technology). S64

## Factors affecting location 10 items

Electric power availability and cost · natural disasters · property tax · corporate income tax (single sales-factor apportionment is more favourable) · sales tax · labour availability and quality · construction cost · telecom infrastructure · economic development incentives · quality of life. S65–S66

International factors: economic, environmental, political and legal, social, technological, plus supply chain proximity, customers, flexibility, language, healthcare and education. S67–S69

## Multi-facility strategies

| | |
|---|---|
| **By product/service** | **P&G**: separate plants for cleaning products and personal care. S70 |
| **By geographic area** | **Amazon**: fulfilment centres for North America, Europe, Asia. S71 |
| **By process** | **Tesla**: Nevada Gigafactory makes batteries; Fremont, California assembles vehicles. S72 |

## Evaluating location alternatives

- **Preliminary screening**: infrastructure, suppliers, costs, quality, free trade zones, political risk, business environment, customer proximity, government barriers. S73
- Qualitative: **simple comparison chart** (e.g. Biratnagar vs Kathmandu on community attitude and lifestyle) and **point rating** (sum the scores; the slide picks B with 34 over A 33 and C 30). S74–S75
- **Factor rating**: list factors → weight them → rating scale → score each site → weighted sum → choose the highest. S76–S78

| FACTOR | WEIGHT | A RATING → SCORE | B RATING → SCORE |
|---|---|---|---|
| **Land cost** | 0.2 | 4 → 0.8 | 3 → 0.6 |
| **Construction cost** | 0.2 | 3 → 0.6 | 5 → 1.0 |
| **Proximity to customers** | 0.15 | 4 → 0.6 | 2 → 0.3 |
| **Proximity to highways** | 0.15 | 5 → 0.75 | 4 → 0.6 |
| **Labour availability** | 0.1 | 4 → 0.4 | 3 → 0.3 |
| **Access to utilities** | 0.1 | 5 → 0.5 | 4 → 0.4 |
| **Total** | | 3.7 (exactly 3.65) | 3.2 |

Factor rating example from S79–S80: Location A wins.

## Simple median model

- Minimizes total transport cost when movement is only along X and Y (rectilinear). S81

| FACILITY | LOCATION (X, Y) | COST/UNIT DISTANCE | ANNUAL LOAD |
|---|---|---|---|
| **Bareilly** | (10, 80) | 10 | 452 |
| **Shahjahanpur** | (30, 60) | 10 | 678 |
| **Gonda** | (80, 50) | 10 | 483 |
| **Kanpur** | (50, 10) | 10 | 711 |
| **Sultanpur** | (80, 10) | 10 | 539 |
| **Total** | | | 2,863 |

1. **Median load** = (n + 1)/2 = (2,863 + 1)/2 = **1,432**. S81
2. **X**: sort by x and accumulate loads: Bareilly 1–452, Shahjahanpur 453–1,130, Kanpur 1,131–1,842 contains 1,432 → **x = 50**. S82
3. **Y**: sort by y: Kanpur and Sultanpur (y = 10) reach 1,250; Gonda (y = 50) reaches 1,733 and contains 1,432 → **y = 50**. New plant at (50, 50). S83
4. Rectilinear distance, e.g. Bareilly: |50 – 10| + |80 – 50| = 70. Cost = distance × load × cost per unit distance. Slide total: **13,46,800**. S83–S84

## Centre of gravity model

- $C_x = \Sigma(w_i x_i) \div \Sigma w_i$; $C_y = \Sigma(w_i y_i) \div \Sigma w_i$; weights are volumes shipped or customers served. S85–S86
- Same data: $x_o$ = (10×452 + 30×678 + 80×483 + 50×711 + 80×539) ÷ 2,863 = **49.66**; $y_o$ = **39.64**. S87
- Location examples: sugar mills near cane farms; tea factories in Assam, West Bengal, Karnataka; ADNOC in the UAE. S88

## Mechanical analogue: Varignon's frame

- Strings through holes in a table carry weights; the common knot **P** settles at the least-cost location (no friction). $R_1 \ldots R_m$ = raw material sources, $M_1 \ldots M_n$ = markets, **m + n** weights; uses **Euclidean** (straight-line) distance; weight = annual trips × cost per unit distance. S89–S90

BEFORE THE EXAM ALL LECTURES

# Cheat sheet

| | |
|---|---|
| **More facilities →** | Inventory ↑, facility cost ↑, response time ↓, transport ↓ then ↑, total logistics cost U-shaped S8–S13 |
| **Outbound vs inbound** | Outbound cost per unit higher (smaller lots) S11 |
| **Six designs** | Mfr direct · in-transit merge · distributor + carrier · distributor + last mile · mfr/distributor pickup · retail pickup S28 |
| **Best response time** | Retail storage with customer pickup (rank 1) S49 |
| **Highest transport cost** | Distributor storage with last-mile delivery (rank 5) S49 |
| **Median model** | Median load = (Σ loads + 1)/2; pick the x and y where cumulative load crosses it S81–S83 |
| **Centre of gravity** | Weighted average of coordinates S86 |
| **Varignon frame** | Physical model, Euclidean distance, m + n weights S90 |

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

Median model = city blocks (rectilinear), centre of gravity = weighted average, Varignon = straight strings (Euclidean).

READ ONCE SOURCE CHECK

# Slide quirks to watch

- **Median model costs:** Bareilly should be 70 × 452 × 10 = 3,16,400 (slide: 3,36,400) and Sultanpur 70 × 539 × 10 = 3,77,300 (slide: 3,77,700). The correct total is 13,26,400; the slide prints 13,46,800. The new location (50, 50) is unaffected. S84
- **Point rating:** the visible B scores (7, 5, 9, 8, 2) add to 31, but the slide shows and circles 34 as the best. Read it as the slide intends (B = 34) if asked. S75
- **Factor rating weights:** the table's weights add to 0.9, not 1, and split market factors as 0.30 and operational as 0.20 (the text says 0.35 and 0.25). Location A sums to 3.65, shown as 3.7. S79–S80
- The cumulative range for Kanpur is printed 1,131–1,842; by the loads it is 1,131–1,841. S82
- "Barely" and "Sharjapur" on S82–S83 mean Bareilly and Shahjahanpur.
- Varignon's frame calls $M_1 \ldots M_n$ "raw markets"; they are market locations. S90

SELF-TEST 41 QUESTIONS

# Week 9 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 42

**Distribution is a key driver of profitability because it directly affects:**

- **A** both supply chain cost and customer value
- **B** only product design
- **C** only taxes
- **D** only marketing spend

Q2 LECTURE 42

**Firms targeting customers who value short response times need:**

- **A** no inventory
- **B** only online sales
- **C** facilities located close to customers
- **D** fewer facilities far away

Q3 LECTURE 42

**'The probability of having a product in stock when a customer order arrives' is:**

- **A** product availability
- **B** product variety
- **C** returnability
- **D** order visibility

Q4 LECTURE 42

**As the number of facilities increases, inventory costs:**

- **A** decrease
- **B** increase
- **C** stay the same
- **D** first increase then decrease

Q5 LECTURE 42

**As the number of facilities increases, transportation cost:**

- **A** decreases until inbound economies of scale are lost, then increases
- **B** always decreases
- **C** is unaffected
- **D** always increases

Q6 LECTURE 42

**Outbound transportation costs per unit tend to be higher than inbound because:**

- **A** outbound trucks are slower
- **B** inbound uses air
- **C** inbound lot sizes are typically larger
- **D** outbound has no economies of scale ever

Q7 LECTURE 42

**In the logistics cost curve, as the number of facilities rises, response time:**

- **A** is constant
- **B** falls
- **C** rises
- **D** is U-shaped

Q8 LECTURE 42

**Partnering with suppliers to ship directly to customers without holding inventory is:**

- **A** decentralized distribution
- **B** drop-shipping
- **C** hybrid distribution
- **D** cross-docking

Q9 LECTURE 42

**Which is listed as a challenge of online sales for customer service?**

- **A** 24/7 access
- **B** Larger product variety
- **C** Order visibility
- **D** Returns are harder and costlier

Q10 LECTURE 42

**Dell assembling PCs only after receiving orders lowers inventory through:**

- **A** postponement
- **B** cross-docking
- **C** aggregation
- **D** drop-shipping

Q11 LECTURE 42

**How many warehouses did Netflix operate from, compared with Blockbuster's thousands of stores?**

- **A** About 500
- **B** About 5,000
- **C** About 50
- **D** About 5

Q12 LECTURE 42

**For non-digital products, aggregating inventory for online sales tends to:**

- **A** reduce outbound transportation costs
- **B** reduce information cost
- **C** increase outbound transportation costs relative to inbound
- **D** eliminate transport cost

Q13 LECTURE 42

**Which is NOT one of the cost factors affected by distribution network design?**

- **A** Inventories
- **B** Transportation
- **C** Facilities and handling
- **D** Advertising

Q14 LECTURE 43

**How many distinct distribution network designs are described?**

- **A** Five
- **B** Four
- **C** Six
- **D** Eight

Q15 LECTURE 43

**Manufacturer storage with direct shipping is also called:**

- **A** customer pickup
- **B** last-mile delivery
- **C** drop-shipping
- **D** cross-docking

Q16 LECTURE 43

**Typical response time for manufacturer storage with direct shipping is:**

- **A** same day
- **B** one hour
- **C** one to two weeks
- **D** one month

Q17 LECTURE 43

**In-transit merge improves on drop-shipping mainly because:**

- **A** returns are free
- **B** the customer receives a single delivery
- **C** no carrier is used
- **D** inventory is held at retail

Q18 LECTURE 43

**In the comparative ranking (1 = strongest), which design ranks 1 for response time?**

- **A** Retail storage with customer pickup
- **B** Manufacturer storage with pickup
- **C** Distributor storage with last-mile delivery
- **D** Manufacturer storage with direct shipping

Q19 LECTURE 43

**Which design ranks worst (6) for facility and handling cost?**

- **A** Manufacturer storage with pickup
- **B** Retail storage with customer pickup
- **C** Manufacturer storage with direct shipping
- **D** Distributor storage with carrier delivery

Q20 LECTURE 43

**Which design has the highest transportation cost (rank 5)?**

- **A** Retail storage with customer pickup
- **B** Distributor storage with last-mile delivery
- **C** Manufacturer storage with in-transit merge
- **D** Distributor storage with carrier delivery

Q21 LECTURE 43

**Manufacturer storage designs rank 1 (strongest) on which attribute?**

- **A** Inventory
- **B** Returnability
- **C** Order visibility
- **D** Response time

Q22 LECTURE 43

**Drop-shipping is most suitable (+2) for which products?**

- **A** Low customer effort
- **B** Quick desired response
- **C** High-demand products
- **D** Very low-demand products

Q23 LECTURE 43

**When did Dell begin distributing through retail stores like Wal-Mart?**

**A** 2002
**B** 2012
**C** 2007
**D** 1997

Q24 LECTURE 43

**Online sales let Dell operate with:**

**A** no IT investment
**B** no suppliers
**C** positive working capital
**D** negative working capital

Q25 LECTURE 43

**For low-cost standardized hardware, Dell's online model rates –2 on:**

**A** response time and transportation
**B** information and facilities
**C** efficient funds transfer
**D** product variety and inventory

Q26 LECTURE 43

**Which company is cited as having struggled with online sales?**

**A** Webvan
**B** Blue Nile
**C** Dell
**D** Amazon

Q27 LECTURE 43

**Which design offers same-day pickup and lets customers inspect the product before buying?**

**A** Retail storage with customer pickup
**B** Distributor storage with carrier delivery
**C** In-transit merge
**D** Manufacturer storage with direct shipping

Q28 LECTURE 44

**Facility location decisions are strategic because they are:**

**A** long-term, difficult to reverse and capital intensive
**B** easy to reverse and cheap
**C** daily decisions
**D** made by suppliers

Q29 LECTURE 44

**The three tiers in the model for assessing location competitiveness are:**

**A** cost, quality, time
**B** supplier, plant, market
**C** local, regional, global
**D** country, sector, company

Q30 LECTURE 44

**Which corporate income tax structure is described as more favourable for plant locations?**

**A** Flat property tax
**B** Property, people and sales apportionment
**C** No tax
**D** Single sales factor apportionment

Q31 LECTURE 44

**Tesla's Gigafactory (batteries) and Fremont factory (assembly) illustrate separate facilities by:**

**A** process
**B** geographic area
**C** customer
**D** product

Q32 LECTURE 44

**In the factor rating example, Location A's total weighted score is about ____ vs B's 3.2.**

**A** 3.2
**B** 3.0
**C** 4.0
**D** 3.7

Q33 LECTURE 44

**The simple median model assumes goods move:**

**A** only along X and Y axes (rectilinear)
**B** by air only
**C** in straight lines
**D** in circles

Q34 LECTURE 44

**With total annual loads of 2,863 units, the median load used is:**

**A** 2,864
**B** 1,432
**C** 1,431.5
**D** 1,431

Q35 LECTURE 44

**In the median model example, the new plant's X coordinate is 50 because:**

**A** 50 is the average
**B** Gonda's cumulative load passes 1,432
**C** Bareilly has the largest load
**D** Kanpur's cumulative range (1,131–1,842) contains the median 1,432

Q36 LECTURE 44

**The optimal location found by the median model in the example is:**

**A** (50, 50)
**B** (49.66, 39.64)
**C** (80, 10)
**D** (30, 60)

Q37 LECTURE 44

**Using rectilinear distance, the distance from Bareilly (10, 80) to the new plant (50, 50) is:**

**A** 50
**B** 70
**C** 30
**D** 40

Q38 LECTURE 44

**The centre of gravity method gives the new facility's X coordinate as:**

**A** $\Sigma x_i \div n$
**B** median of $x_i$
**C** max $x_i$
**D** $\Sigma(x_i d_i) \div \Sigma d_i$

Q39 LECTURE 44

**Using the slide's data, the centre of gravity location is about:**

**A** (49.66, 39.64)
**B** (45, 45)
**C** (80, 50)
**D** (50, 50)

Q40 LECTURE 44

**Varignon's frame finds the least-cost location using which distance measure?**

**A** Network
**B** Rectilinear
**C** Manhattan only
**D** Euclidean (straight line)

ANSWERS 41 QUESTIONS

# Answer key

| Q | ANS | WHY · SLIDE |
|---|---|---|
| Q1 | A | S3. S3 |
| Q2 | C | S5; S8 shows required facilities rise as desired response time falls. S5 |
| Q3 | A | S5. S5 |
| Q4 | B | S7 and S9. S9 |
| Q5 | A | U-shaped curve. S10–S11 |
| Q6 | C | Amazon receives full truckloads but ships small packages. S11 |
| Q7 | B | Total logistics cost is U-shaped. S13 |
| Q8 | B | S15. S15 |
| Q9 | D | S20. S20 |
| Q10 | A | Amazon printing low-volume books on demand is another example. S21 |
| Q11 | C | S22. S22 |
| Q12 | C | S23. S23 |
| Q13 | D | Inventories, transportation, facilities and handling, information. S7 |
| Q14 | C | S28. S28 |
| Q15 | C | S30. S30 |
| Q16 | C | Because of distance and two stages of order processing. S32 |
| Q17 | B | Better customer experience; somewhat lower transport cost. S35 |
| Q18 | A | Last-mile delivery ranks 2. S49 |
| Q19 | B | Retail stores are expensive facilities. S49 |
| Q20 | B | S42: very high cost given minimal scale economies. S49 |
| Q21 | A | Aggregation at the manufacturer gives the lowest inventory; also variety and availability. S49 |
| Q22 | D | Retail pickup is +2 for high-demand products. S50 |
| Q23 | C | S51. S51 |
| Q24 | D | Dell collects payment quickly. S52 |
| Q25 | A | Customized hardware gets −1 on both. S54 |
| Q26 | A | S55. S55 |
| Q27 | A | S46. S46 |
| Q28 | A | S60. S60 |
| Q29 | D | S64. S64 |
| Q30 | D | S65. S65 |
| Q31 | A | P&G: by product; Amazon: by geography. S72 |
| Q32 | D | 0.8 + 0.6 + 0.6 + 0.75 + 0.4 + 0.5 = 3.65, shown as 3.7. A is chosen. S80 |
| Q33 | A | S81. S81 |
| Q34 | B | (n + 1)/2 = 2,864/2 = 1,432. S81 |
| Q35 | D | S82. S82 |
| Q36 | A | (49.66, 39.64) is the centre of gravity answer. S83 |
| Q37 | B | \|50 − 10\| + \|80 − 50\| = 40 + 30 = 70. S83 |
| Q38 | D | Weighted average using loads $d_i$. S87 |
| Q39 | A | $x_o$ = 142,170 ÷ 2,863 = 49.66; $y_o$ = 39.64. S87 |
| Q40 | D | The knot P comes to rest at the least-cost point (no friction). S90 |
| Q41 | B | Each weight = annual trips × cost per unit distance. S90 |

Made from *WEEK 9.pdf* (93 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except boxes marked "my own".